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3-1: Take Out What They Share

TEKS P.5.J: solve polynomial equations by applying a variety of techniques. Factoring out the greatest common factor is the first tool.

Warm-up: Build it, then undo it

Multiply (distribute): (a) $6(2x + 1)$ = _____ (b) $3x(x - 4)$ = _____ (c) $5x^2(x + 2)$ = _____

Now run it backward. Fill in the blanks: (d) $12x + 6$ = _____(_____ + _____) (e) $10x^2 - 15x$ = _____(_____ – _____)

(f) In your own words, what is a *greatest common factor*?

Part A: Find the GCF

Find the GCF of the numbers, then of the variable parts, then put them together. The first row is done for you.

Terms	GCF of the numbers	GCF of the variable parts	GCF of the whole
18 and 24	6	(none)	6
$12x^2$ and $8x^3$			
$15x^4$ and $25x$			
$4x^2y^3$ and $10xy^2$			
$9x$ and $18x^2$			

When a variable is in every term, the GCF uses the _____ exponent it shows. Why that one?

Part B: Did the 1 disappear?

Two students factored **$7x^3 + 7x^2$**. **Maya:** $7x^2(x)$ **Jon:** $7x^2(x + 1)$

(a) Distribute each answer back. Who matches $7x^3 + 7x^2$? (b) What do you multiply $7x^2$ by to get $7x^2$? (c) Finish: when a term *is* the GCF, what is left in the parentheses?

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Practice

Factor out the greatest common factor. Check each by distributing. (B) = Basic target (factor out the GCF); "stretch" goes a bit beyond.

1. (B) $18x + 24$

2. (B) $x^5 - 7x^2$

3. (B) $8x^3 + 20x^2$

4. (B) $9x^4 - 12x^3 + 21x^2$

5. (B) $4x^2y^3 + 10xy^2$

6. (B) $-6x^2 - 15x$ (Take out a negative GCF.)

7. (B) $6x^2 + 6x$

8. (B) $4x^4 + 12x^3 + 4x^2$

9. (B) *stretch:* $3(x + 2) + x(x + 2)$ (The shared piece is a binomial.)

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10. (B) *stretch:* $2x(x + 3) - (x + 3)$

11. (B) *stretch:* Solve $\mathbf{3x^2 = 12x}$.

(a) Move everything to one side, factor out the GCF, and set each factor equal to 0.

(b) Alex divided both sides by 3x and got x = 4. Which solution did Alex lose? Why is dividing by x risky?

12. (B) A ball is tossed straight up. Its height in feet after *t* seconds is $\mathbf{h(t) = -16t^2 + 48t}$.

(a) Factor out the GCF, including the negative.

(b) Solve h(t) = 0.

(c) What do your two answers mean for the ball?

13. (B) A rectangular banner has area $\mathbf{8x^2 + 12x}$ square feet. Its width is 4x feet.

(a) Factor $8x^2 + 12x$. (b) Use (a) to write the length. (c) If x = 3, find the width, the length, and the area. Check the area with $8x^2 + 12x$.

Math Time Machine: Around 300 BCE, Euclid wrote down a step-by-step method for finding the greatest common measure of two numbers, often called the oldest algorithm still in common use. He also pictured the distributive law as rectangles whose areas add up. And the word **factor** comes from the Latin *facere*, "to make."